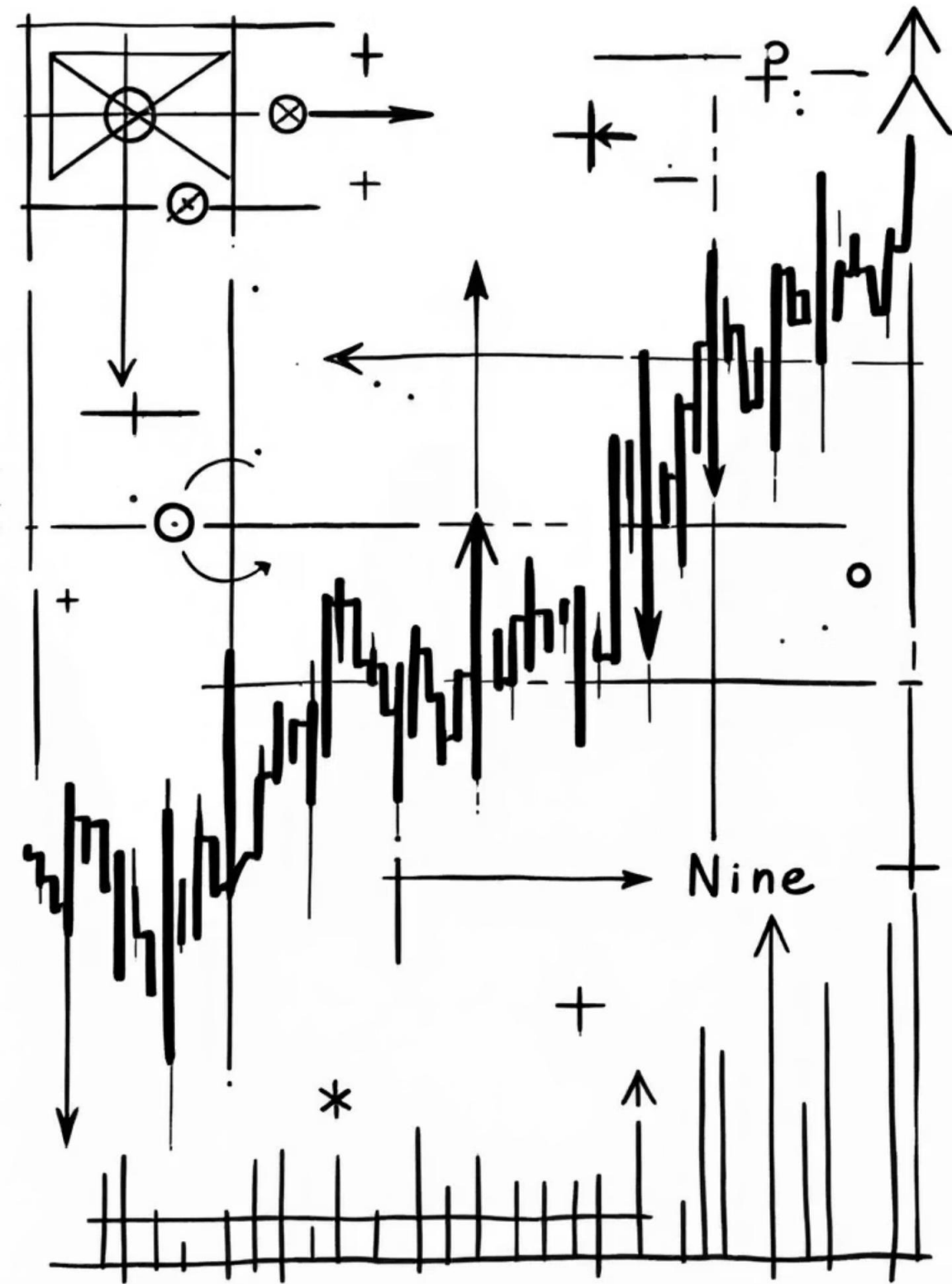


Assessing Gann Square Effectiveness in Machine Learning-Based Stock Predictions

A thesis on exploring whether W.D. Gann's century-old geometric trading methods can enhance modern machine learning models for stock market prediction and trading.



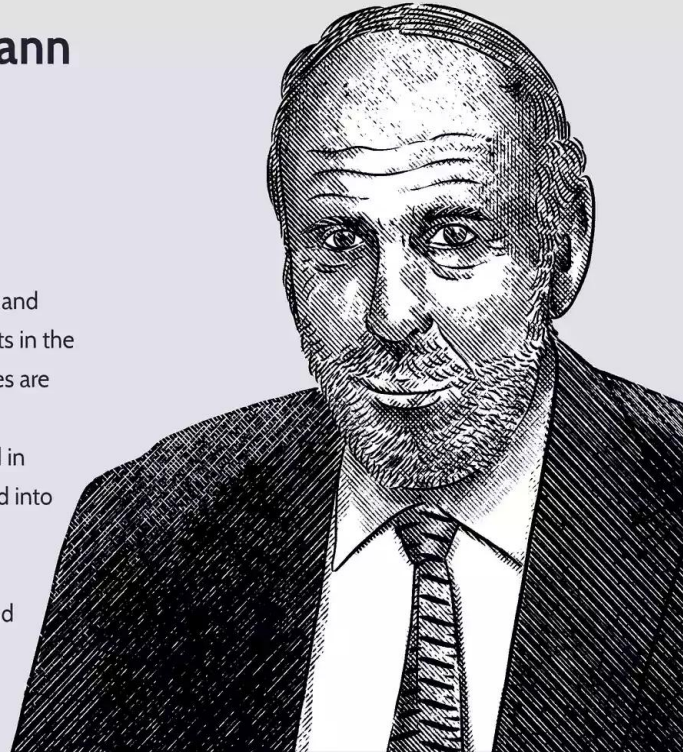
William Delbert Gann

Born: June 6, 1878

Died: June 18, 1955

Trader

- Known for using geometry, astrology and ancient mathematics to predict events in the financial markets; his trading strategies are still widely used today
- Some of his ideas on trading are used in technical analysis tools and integrated into current charting and trading software platforms
- His trading theories have doubters and supporters, alike



Who is W.D. Gann?

Gann's Core Premises

- Price, time, and pattern are key factors in market behavior
- Markets move in repeating cycles
- Price movements exhibit geometric relationships

Square of Nine

A numerical spiral chart where natural numbers are arranged clockwise, creating geometric alignments. Used to derive support and resistance levels that could act as future turning points.

Gann Angles

Straight lines drawn on price-time charts at specific fixed slopes (1×1 , 2×1 , etc.). They separate price action into zones of trend strength and provide continuously updating references for support/resistance.

Time Cycles

Gann believed specific durations often precede market reversals. Traders measure days between significant turning points and project those lengths forward to anticipate the next ones.

Stock Prediction Current progress

Complex Market Dynamics

Stock markets are notoriously volatile and complex, driven by myriad factors that create non-linear, often noisy price movements.

Modern AI Approaches

Machine learning techniques have improved predictive performance compared to naive methods, but achieving consistently high accuracy remains challenging.

Gann's Geometric Methods

W.D. Gann (1878-1955) developed unique geometric techniques like the Square of Nine and Gann Angles to predict market movements based on price-time relationships.

This research investigates whether Gann's controversial methods, often compared to "financial horoscopes," can provide valuable signals when integrated with modern machine learning and reinforcement learning approaches.

Research Questions

1 Prediction Accuracy

Can models augmented with multiple Gann-based features outperform standard machine learning models in terms of error metrics (RMSE, MAE, MAPE)?

2 Comparison with Technical Indicators

Do Gann Square-derived support and resistance levels improve prediction accuracy over traditional technical indicators (such as moving averages or RSI)?

3 Trading Performance

Does including Gann Square of Nine support/resistance levels as features improve the trading performance of a trading agent relative to a baseline agent using only price data?

4 Technical Indicators vs. Gann Features

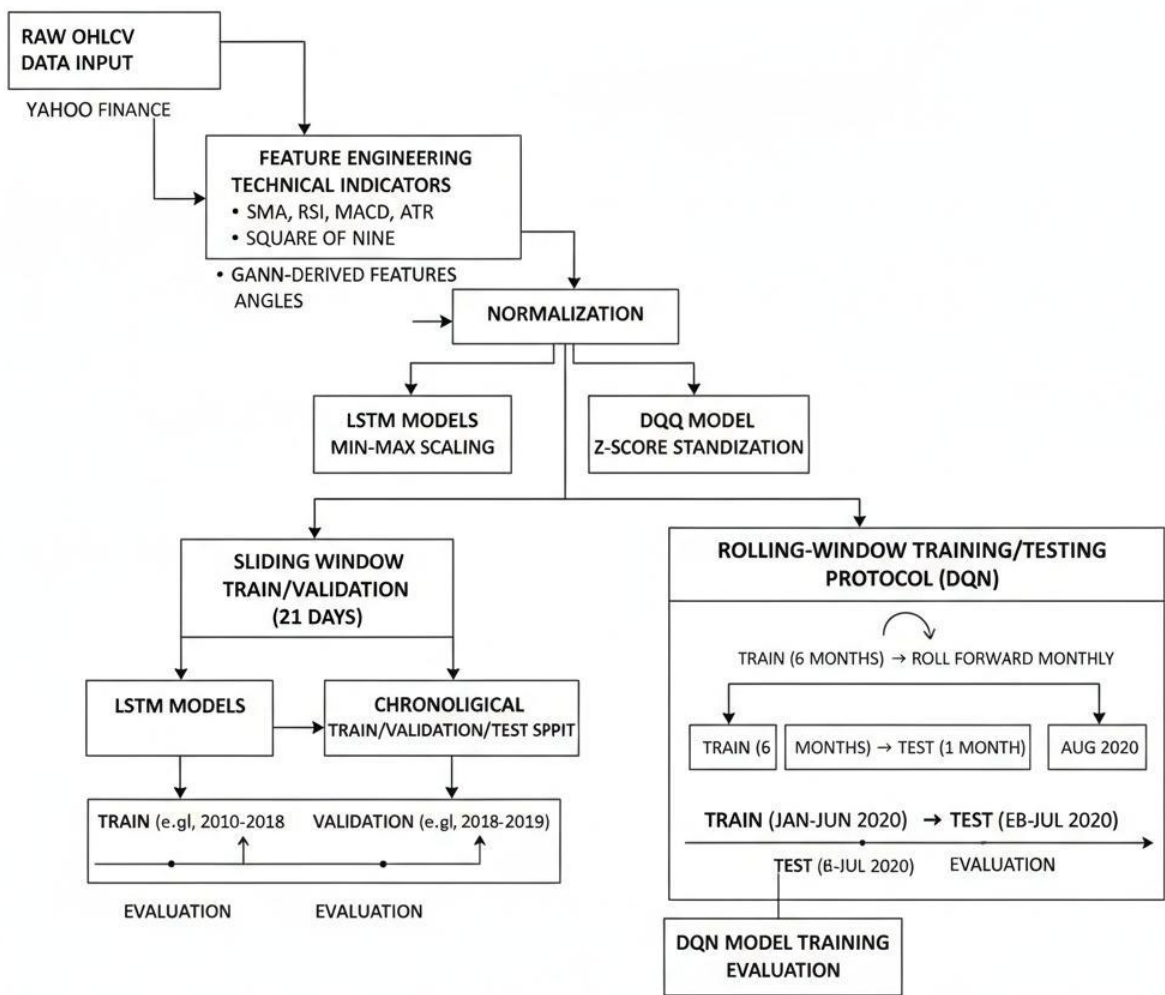
How does adding standard technical indicators affect the performance of trading agents compared to Gann-based features?

5 Gann Angle Enhancement

Does incorporating Gann angle features further enhance agent performance beyond the Gann Square features?

Data and Preprocessing

STOCK MARKET PREDICTION PIPELINE: DATA PREPROCESSING & TRAINING/TESTING PROTOCOL



Data Sources

Daily OHLCV data from Yahoo Finance (2011-2023):

- Microsoft (MSFT)
- Apple (AAPL)
- Google (GOOG)
- Nifty 50 (NSEI)

Preprocessing Steps

- Normalization using Min-Max scaling
- 21-day sliding windows for LSTM inputs
- Chronological splitting (no shuffling)
- Training: 70%, Validation: 15%, Testing: 15%

Gann-Derived Features

Gann Square of Nine:

- Resistance = $(\sqrt{P} + 1)^2$
- Support = $(\sqrt{P} - 1)^2$

Where P is previous day's closing price

Gann Angles:

- LSTM: 100-day rolling low as pivot
- DQN: Dynamic pivot-based swing highs/lows

Methodology: LSTM

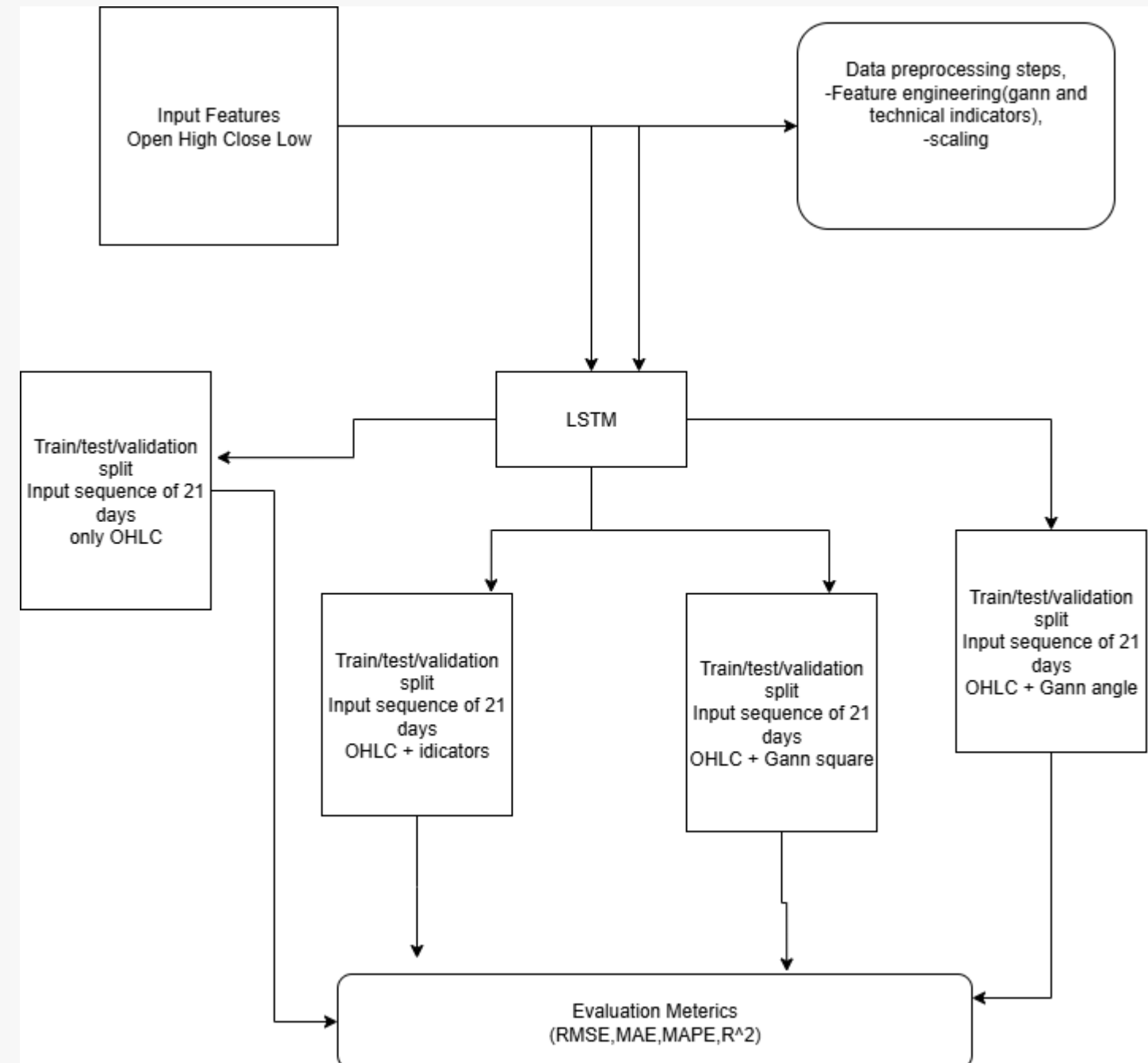
LSTM for Prediction

Long Short-Term Memory networks to forecast next-day closing prices using 21-day sliding windows of historical data.

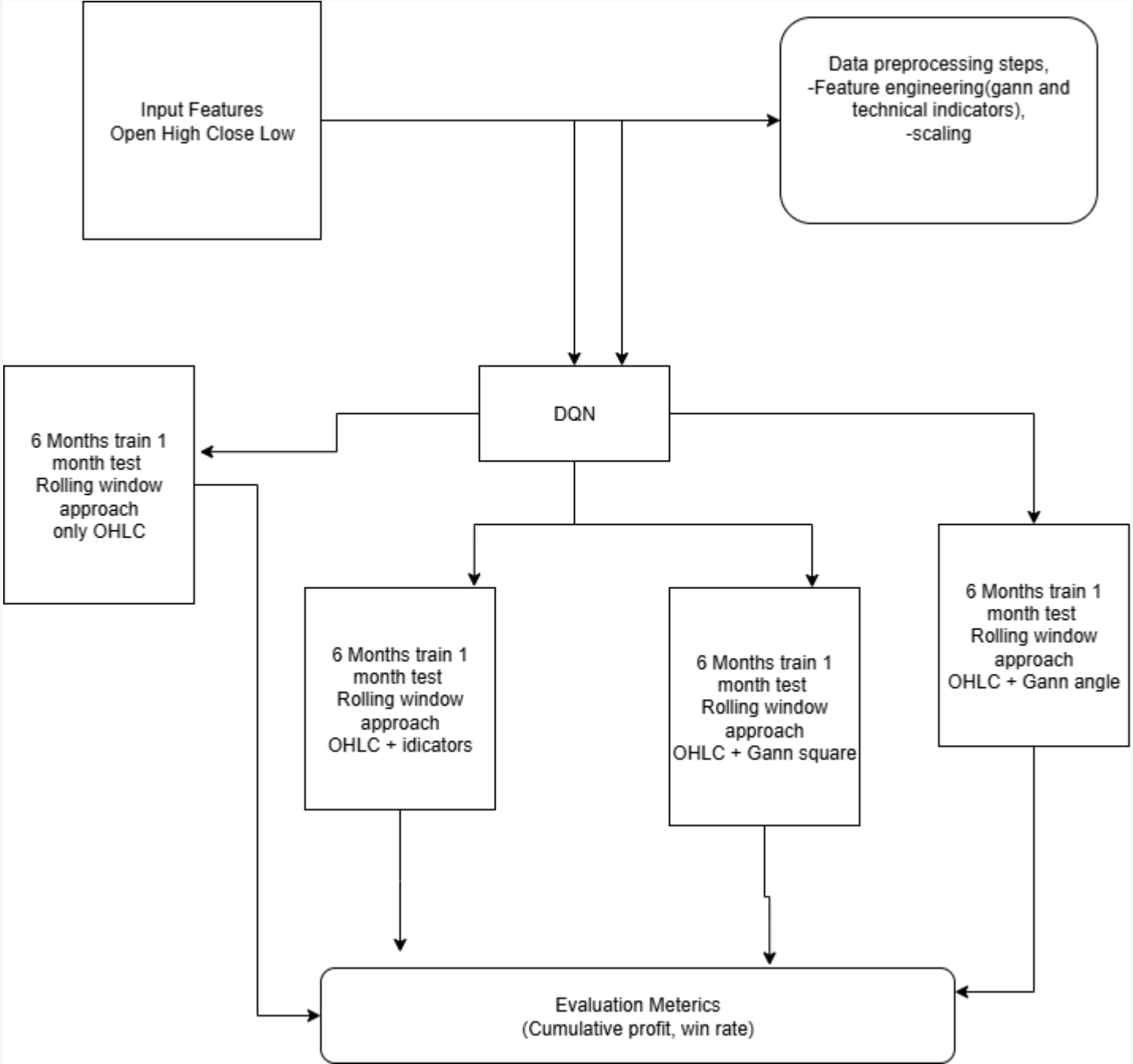
Feature Sets Tested

Baseline: Raw OHLC (Open, High, Low, Close) values only

Gann Square: OHLC with support and resistance levels from Square of Nine



Methodology: DQN



DQN for Trading

Deep Q-Network reinforcement learning agents to make buy/sell/hold decisions and maximize returns.

Technical Indicators: OHLC with RSI, MACD, moving averages, and ATR

Gann Angle: OHLC with angle-based features recalculated using rolling lookback

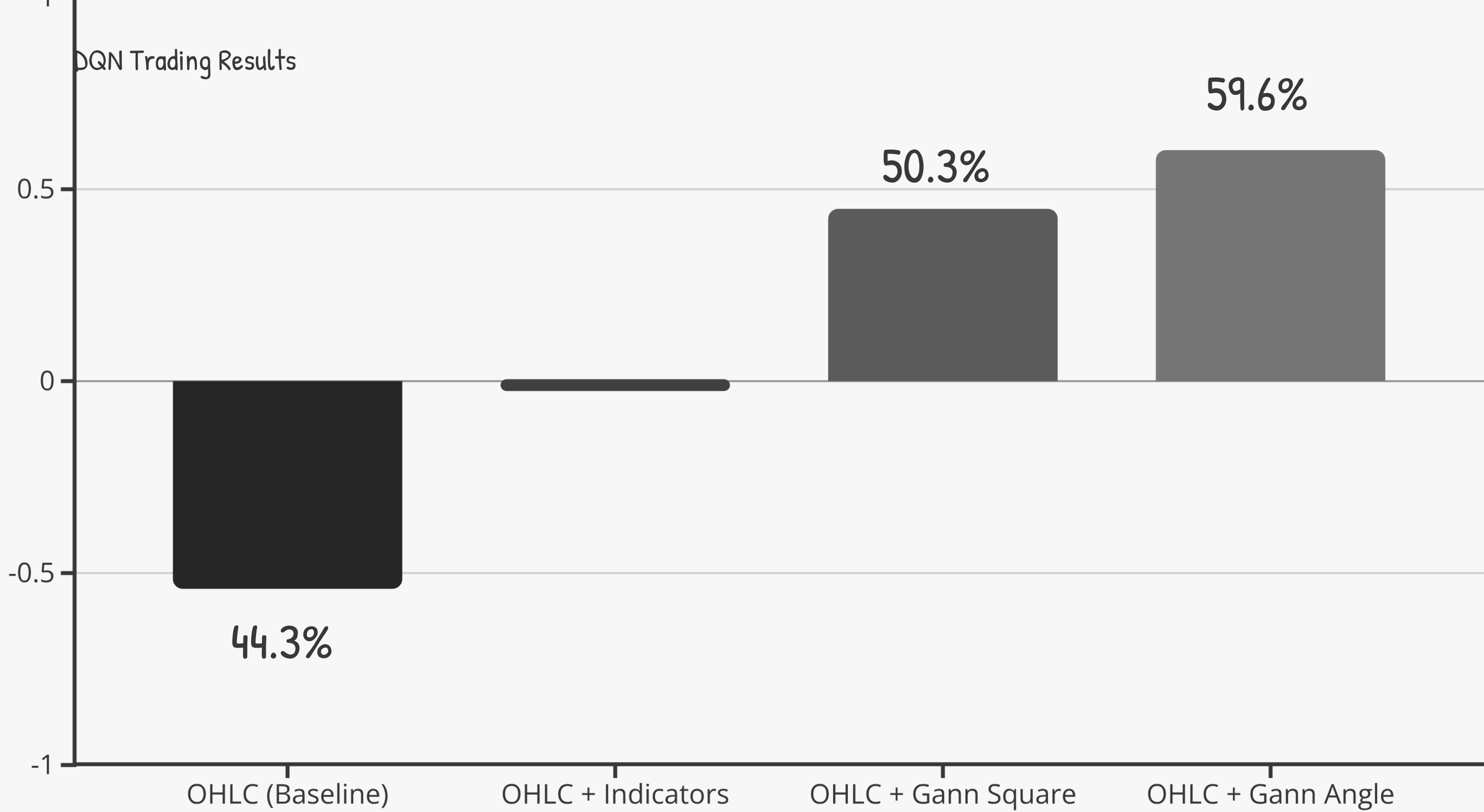
LSTM Prediction Results

Performance across different feature sets for stock price prediction:

Features	RMSE	MAE	MAPE	R²
OHLC (Baseline)	6.64	5.20	0.0253	0.93
OHLC + Indicators	23.98	19.52	0.0972	0.20
OHLC + Gann Square	8.82	6.88	0.0327	0.87
OHLC + Gann Angle	17.74	14.31	0.0673	0.49

The baseline OHLC models consistently achieved the lowest errors. Adding technical indicators or Gann features generally degraded performance. This suggests that Gann-derived features do not improve predictive accuracy for exact price forecasting.

DQN Trading Results



Discussion: Why the Difference?

Why Gann Failed in Prediction

- Gann methods are geometric rather than statistical
- Closing price data is very noisy
- Gann features weren't designed to predict exact prices
- Patterns may already be encoded in OHLC sequences

Why Gann Succeeded in Trading

- RL optimizes for decision-making, not prediction accuracy
- Gann features highlight potential reversal zones
- Trading agents benefit from knowing when to act
- Gann signals helped capture larger positive moves

The key insight: Gann features may not provide strong predictive power for exact prices, but in a decision-making context they offer situational value for identifying potential turning points.

Conclusion and Future Work

Key Findings

- Gann features do not improve predictive accuracy
- Gann features do improve trading performance
- What fails as a predictor can succeed as a decision-support signal

Limitations

- Limited dataset scope (only 4 stocks)
- Daily data only (no intraday testing)
- Lightweight model architectures
- Basic evaluation metrics

Future Work

- Expand to more stocks and asset classes
- Test with intraday/minute-level data
- Explore more sophisticated architectures
- Include risk-adjusted performance metrics

This research contributes to the ongoing debate on the relevance of traditional trading methods in the age of artificial intelligence. The results show that Gann-based features add little value in predictive modeling but may provide situational advantages in trading agents.